

Reese Ludwick

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SUMMARY

AI/ML engineer who **builds and owns production MCP servers** and **video/creative-understanding AI pipelines**. Ship custom FastMCP servers and external MCP integrations giving agents dynamic tool access in production, plus a full video-understanding pipeline (autoclip.studio) that transcribes, embeds, and learns a creator's style into a scoring "DNA" module. Strong on **agentic workflows** (LangGraph multi-agent), **model-evaluation** loops, embeddings/retrieval (Voyage AI + pgvector), and FastAPI production services.

CORE SKILLS

MCP & Tooling: Custom FastMCP servers, external MCP integrations, dynamic agent tool access, AWS-deployed
Video / Creative Understanding: WhisperX transcription, Voyage AI embeddings, pgvector, ffmpeg rendering, learned style/"DNA" scoring module

Agentic & Eval: LangGraph multi-agent (codegen/testing/security/supervision), LangChain, RAG, model evaluation, prompt engineering, NLP clustering, executive reporting

Backend & Data: Python, FastAPI, PostgreSQL, Redis, Celery, SQL, Django REST

ML & Platform: Voyage AI, Sentence Transformers, TensorFlow, Pandas, NumPy, Amazon Bedrock, Anthropic API, Docker, Git/GitHub, Linux/Ubuntu, Azure Fundamentals, React/TypeScript, Claude Code

EXPERIENCE

Software Associate

Remote

Cognizant

Dec 2025–Present

- Build GenAI and agentic AI systems with LangChain, LangGraph, and Python.
- **Own MCP server integrations** connecting enterprise tools and LLMs for production workflows; build RAG pipelines behind them.
- Scope, prototype, and deliver AI data tooling cross-functionally.

GenAI Developer Contractor

Remote

Cognizant

Jun–Dec 2025

- **Built custom FastMCP servers and deployed external MCP integrations on AWS** to give agents dynamic tool access — the core capability this role centers on.
- Ran **prompt engineering and model evaluation** to optimize LLM outputs across GenAI POCs.
- Shipped modular, production-style agentic workflows.

Freelance AI Engineer & Full-Stack Developer

Remote

WorldCovers / APMC

May 2026–Present

- Shipped production web properties on a 14-table PostgreSQL schema (Django REST + React/TypeScript).
- AI-augmented engineering (Claude Code) on compressed timelines with direct stakeholder coordination.

PROJECTS

CreatorClip / autoclip.studio — *creative + video understanding, scoring*

Python, FastAPI, Celery, pgvector, WhisperX, Voyage AI, Anthropic API, ffmpeg, YouTube OAuth

- AI video-clipping pipeline that **learns a creator's channel style** via a custom **"DNA" scoring module** — a direct analog to scoring *why creatives work*.
- Async Celery pipeline: WhisperX transcription → Voyage AI + pgvector embeddings → ffmpeg rendering; full YouTube OAuth.
- Live SaaS (Stripe billing) that turns raw footage into structured, embedded, scoreable content understanding.

Agentic Software Development Team — *agentic workflows + evaluation*

LangChain, LangGraph, Amazon Bedrock (Cognizant R&D)

- Multi-agent LangGraph system (codegen / testing / security / supervision) with dedicated **evaluation and security-review agents**; Bedrock + OSS ML backends.

Customer Call Data Analysis & Insights Engine — *eval/scoring over unstructured data*

Python, NLP, Sentence Transformers (Cognizant R&D)

- NLP clustering + embeddings grouping unstructured call data into semantic segments; agentic analysis feeding executive reporting.

CFO Agent — *agentic financial assistant (in active development)*

Python, FastAPI, LangGraph, PostgreSQL, Redis, Anthropic API

- FastAPI + LangGraph agent with PostgreSQL persistence, Redis caching, CSV/OFX import + encrypted vault; hybrid context architecture; multi-tenant-ready.

EDUCATION

BS, Computer Science — Shepherd University · Aug 2021–May 2025 · Major GPA 3.43

CERTIFICATIONS

TensorFlow Developer Specialization · Linux + Git (Boot.dev) · AZ-900 · GitHub Copilot · GenAI Context Engineering (Cognizant)